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Original practice paper

JAMB UTME Mathematics: Practice Paper 1

Twenty original questions in the style of the UTME, with answers and worked explanations.

Exam	Subject	Questions	Time
JAMB UTME	**Mathematics**	**20**	**25 minutes**

Instructions

- Answer every question. Each has one correct option.
- Calculators are not allowed, as in the real UTME.
- Time yourself: aim for just over a minute per question.
- Check your answers on the last pages when you finish.

1. Simplify $8^{2/3} \times 4^{-1/2}$.

A 2
B 4
C 1/2
D 8

2. Given that $\log_{10} 2 = 0.3010$, find $\log_{10} 8$.

A 0.6020
B 0.9030
C 2.4080
D 0.0903

3. Find the roots of $x^2 - 7x + 12 = 0$.

A x = 3 or x = 4
B x = −2 or x = 6
C x = 2 or x = 6
D x = −3 or x = −4

4. Convert 1011_2 to a base ten number.

A 9
B 10
C 11
D 13

5. Find the 10th term of the arithmetic progression 3, 7, 11, ...

A 43
B 36
C 39
D 40

6. Find the sum of the first four terms of the geometric progression 2, 6, 18, ...

A 162
B 54
C 78
D 80

7. A bag contains 4 red balls and 6 blue balls. One ball is picked at random. What is the probability that it is red?

A 3/5
B 2/5
C 2/3
D 1/4

8. Find the mean of 4, 7, 9, 10 and 15.

A 10
B 7
C 8
D 9

9. Find the median of 3, 8, 5, 12, 7 and 10.

A 7
B 8
C 6.5
D 7.5

10. y varies directly as x, and y = 12 when x = 4. Find y when x = 7.

A 3
B 21
C 16
D 28

11. Find the simple interest on ₦20,000 at 5% per annum for 3 years.

A ₦3,000
B ₦3,150
C ₦1,000
D ₦30,000

12. What is the sum of the interior angles of a hexagon?

A 360°
B 720°
C 1080°
D 540°

13. Find the area of a circle of radius 7 cm. (Take π = 22/7.)

A 308 cm²
B 154 cm²
C 44 cm²
D 22 cm²

14. If sin θ = 3/5 and θ is acute, find cos θ.

A 5/3
B 5/4
C 3/4
D 4/5

15. Find dy/dx if $y = 3x^2 - 5x + 2$.

A 3x − 5
B 6x − 5
C 6x + 2
D $6x^2 - 5$

16. n(A) = 12, n(B) = 9 and n(A ∩ B) = 4. Find n(A ∪ B).

A 13
B 17
C 21
D 25

17. Simplify 2/3 ÷ 4/9.

A 8/27
B 6/7
C 3/2
D 2/3

18. Solve the simultaneous equations 2x + 3y = 12 and x − y = 1. Find x.

A 1
B 4
C 2
D 3

19. Find the gradient of the line joining (2, 3) and (6, 11).

A 8
B 2
C 1/2
D 4

20. A trader buys an item for ₦4,000 and sells it for ₦5,000. Find the percentage profit.

A 20%
B 30%
C 10%
D 25%

Answers and explanations

Mark your paper first, then read the explanation for every question you missed.

1. **A**	2. **B**	3. **A**	4. **C**	5. **C**
6. **D**	7. **B**	8. **D**	9. **D**	10. **B**
11. **A**	12. **B**	13. **B**	14. **D**	15. **B**
16. **B**	17. **C**	18. **D**	19. **B**	20. **D**

1. A. $8^{2/3} = 4$ and $4^{-1/2} = 1/2$, so $4 \times 1/2 = 2$.

2. B. $\log 8 = 3 \log 2 = 3 \times 0.3010 = 0.9030$.

3. A. $(x - 3)(x - 4) = 0$, so $x = 3$ or $x = 4$.

4. C. $1\times8 + 0\times4 + 1\times2 + 1\times1 = 11$.

5. C. $a = 3$, $d = 4$, so $T_{10} = 3 + 9 \times 4 = 39$.

6. D. $2 + 6 + 18 + 54 = 80$.

7. B. 4 of the 10 balls are red: $4/10 = 2/5$.

8. D. The sum is 45 and there are 5 numbers, so the mean is 9.

9. D. In order: 3, 5, 7, 8, 10, 12. The median is $(7 + 8) \div 2 = 7.5$.

10. B. $y = kx$ gives $k = 3$, so $y = 3 \times 7 = 21$.

11. A. $I = PRT/100 = 20{,}000 \times 5 \times 3 \div 100 =$ ₦3,000.

12. B. $(n - 2) \times 180° = 4 \times 180° = 720°$.

13. B. $\pi r^2 = 22/7 \times 49 = 154$ cm².

14. D. A 3-4-5 triangle gives an adjacent side of 4, so $\cos \theta = 4/5$.

15. B. Differentiate term by term: $6x - 5$.

16. B. $12 + 9 - 4 = 17$.

17. C. $2/3 \times 9/4 = 18/12 = 3/2$.

18. D. $x = y + 1$, so $5y + 2 = 12$, $y = 2$ and $x = 3$.

19. B. $(11 - 3) \div (6 - 2) = 8 \div 4 = 2$.

20. D. Profit is ₦1,000 on a cost of ₦4,000: $1{,}000 \div 4{,}000 \times 100 = 25\%$.